Audio Captioning: Systematic Ablation Study

Pipeline: CLAP audio encoder (frozen) -> MLP projection -> Language Model
Dataset: MusicCaps (~4,500 train / ~500 val / ~550 test)
Baseline: 2-layer MLP, prefix_len=8, dropout=0.3, greedy decoding, LRs tuned per model (Optuna, 16 trials)
Noise floor: 3 seeds per model (42, 43, 44) -- deltas below 1 std are not interpreted as effects

Model	Type	Params	S1 Epochs	Batch
GPT-2	Decoder-only	124M	30	8
T5-base	Encoder-decoder	220M	50	8
Llama	Decoder-only	1B	40	4

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Loaded 23 arch + 75 decoding eval results, 47 training results, 2 noise floors
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1. Noise Floor (Run-to-Run Variance)

Measured via 3 training seeds (42, 43, 44) per model on identical test set.
Bold = delta > 2 std (likely real). *Grey* = delta < 1 std (within noise).

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  GPT2 -- Noise Floor (3 seeds)
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seed	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	SPICE	SPIDEr	FENSE
42	0.309100	0.063100	0.126500	0.232900	0.110900	0.109100	0.110000	0.399800
43	0.312600	0.070200	0.133800	0.236100	0.125400	0.102700	0.114100	0.409500
44	0.309700	0.069200	0.133200	0.235600	0.122000	0.106400	0.114200	0.382400
std	**0.001900**	**0.003900**	**0.004100**	**0.001700**	**0.007600**	**0.003200**	**0.002400**	**0.013700**

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  T5 -- Noise Floor (3 seeds)
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seed	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	SPICE	SPIDEr	FENSE
42	0.300300	0.057400	0.116200	0.227500	0.084700	0.086600	0.085700	0.395400
43	0.316400	0.066300	0.123400	0.231400	0.096100	0.103200	0.099600	0.379000
44	0.292600	0.065100	0.120100	0.232800	0.120100	0.110800	0.115500	0.445200
std	**0.012200**	**0.004800**	**0.003600**	**0.002700**	**0.018100**	**0.012400**	**0.014900**	**0.034500**

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LLAMA: noise floor not yet computed
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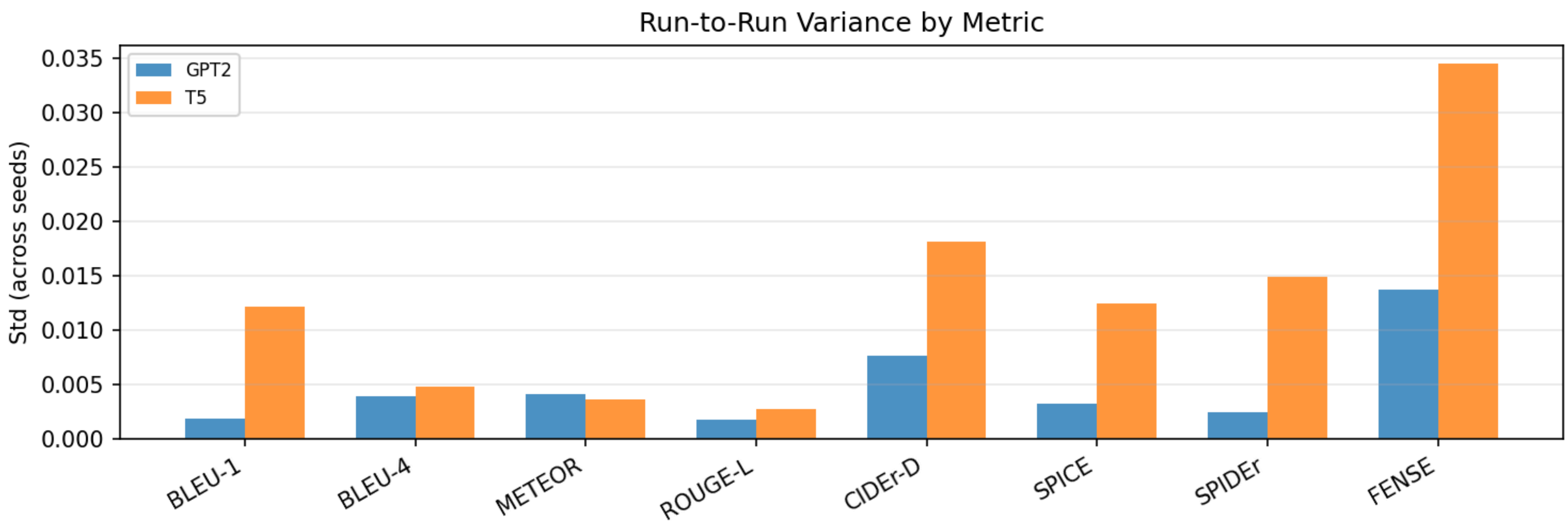
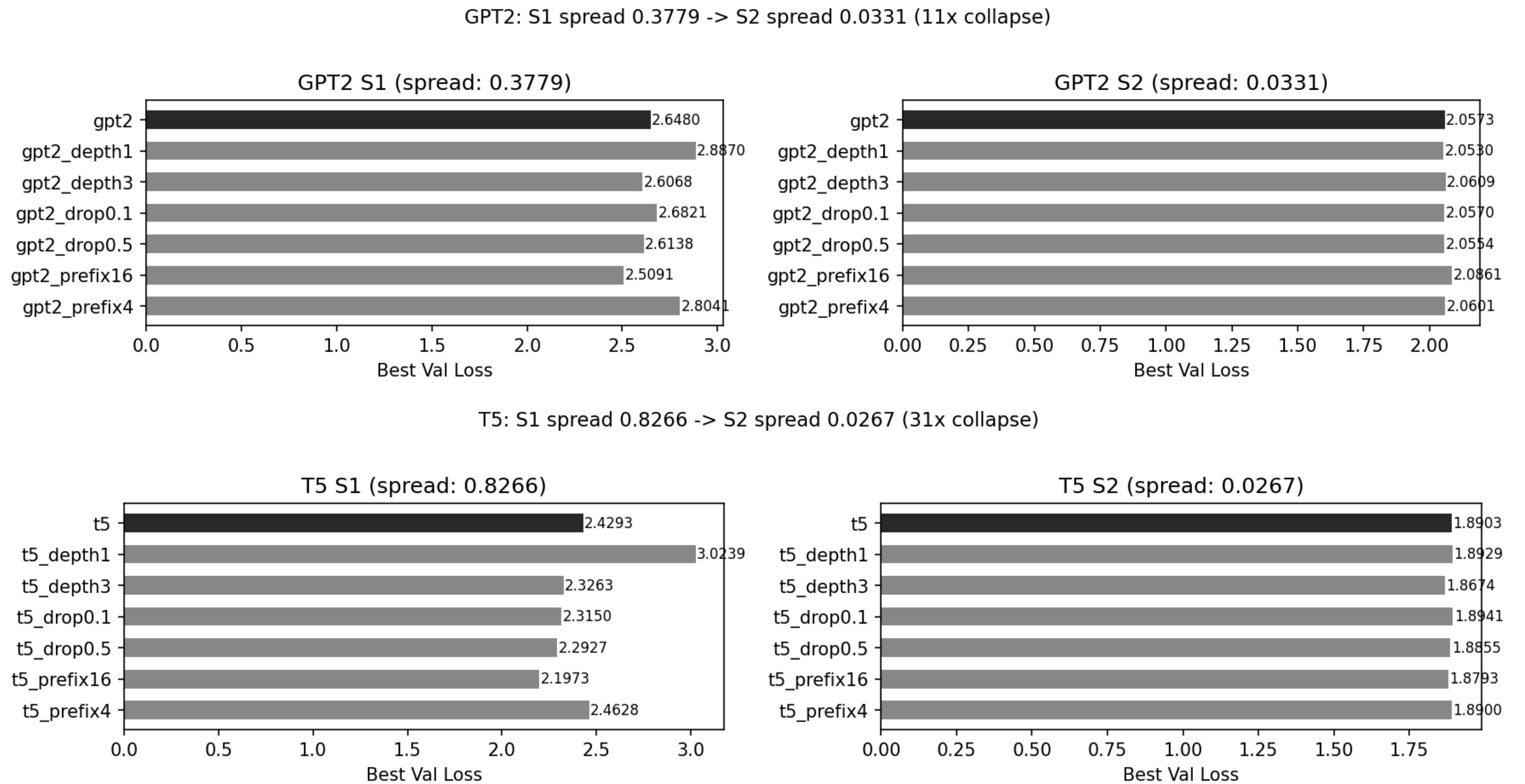


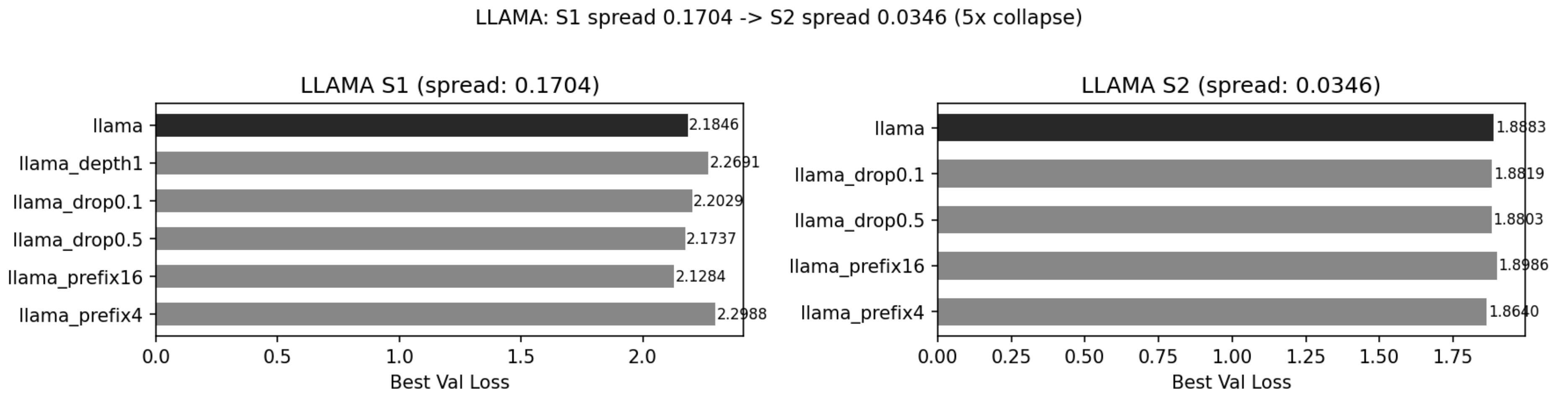
2. Baseline Comparison + S1/S2 Val Loss Spread

S1 trains only the projection; S2 fine-tunes the full LM.
S2 collapses the S1 val loss spread -- final model quality should be judged by test-set captioning metrics, not val loss.

Baseline Test-Set Metrics

model	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	SPICE	SPIDEr	FENSE
gpt2	0.309100	0.063100	0.126500	0.232900	0.110900	0.109100	0.110000	0.399800
t5	0.300300	0.057400	0.116200	0.227500	0.084700	0.086600	0.085700	0.395400
llama	0.310400	0.066900	0.131000	0.236500	0.117000	0.115600	0.116300	0.362600


GPT2: S1 spread 0.3779 -> S2 spread 0.0331 (11x collapse)
GPT2 S1 (spread: 0.3779)
gpt2 2.6480
gpt2_depth1 2.8870
gpt2_depth3 2.6068
gpt2_drop0.1 2.6821
gpt2_drop0.5 2.6138
gpt2_prefix16 2.5091
gpt2_prefix4 2.8041
0.0 0.5 1.0 1.5 2.0 2.5 3.0
Best Val Loss
GPT2 S2 (spread: 0.0331)
gpt2 2.0573
gpt2_depth1 2.0530
gpt2_depth3 2.0609
gpt2_drop0.1 2.0570
gpt2_drop0.5 2.0554
gpt2_prefix16 2.0861
gpt2_prefix4 2.0601
0.00 0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00
Best Val Loss
T5: S1 spread 0.8266 -> S2 spread 0.0267 (31x collapse)
T5 S1 (spread: 0.8266)
t5 2.4293
t5_depth1 3.0239
t5_depth3 2.3263
t5_drop0.1 2.3150
t5_drop0.5 2.2927
t5_prefix16 2.1973
t5_prefix4 2.4628
0.0 0.5 1.0 1.5 2.0 2.5 3.0
Best Val Loss
T5 S2 (spread: 0.0267)
t5 1.8903
t5_depth1 1.8929
t5_depth3 1.8674
t5_drop0.1 1.8941
t5_drop0.5 1.8855
t5_prefix16 1.8793
t5_prefix4 1.8900
0.00 0.25 0.50 0.75 1.00 1.25 1.50 1.75
Best Val Loss



3. Architecture Ablations (Test-Set Metrics)

Baseline: depth=2, prefix_len=8, dropout=0.3, greedy decoding.

Each variant changes one variable. Baseline row shows mean +/- std from seed replication. **Bold** = best per column.

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GPT2 -- Architecture Ablations
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tag	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	SPICE	SPIDEr	FENSE
gpt2	0.3091 ±0.0019	0.0631 ±0.0039	0.1265 ±0.0041	0.2329 ±0.0017	0.1109 ±0.0076	0.1091 ±0.0032	0.1100 ±0.0024	0.3998 ±0.0137
gpt2_depth1	0.3103	0.0631	0.1288	0.2339	0.1078	**0.1164**	0.1121	0.3992
gpt2_depth3	0.2951	0.0600	0.1255	0.2201	0.0952	0.1000	0.0976	0.4002
gpt2_drop0.1	**0.3106**	0.0696	**0.1337**	**0.2371**	**0.1268**	0.1152	**0.1210**	0.4092
gpt2_drop0.5	0.3052	0.0653	0.1297	0.2317	0.1080	0.1050	0.1065	**0.4316**
gpt2_prefix16	0.3093	**0.0698**	0.1324	0.2345	0.1235	0.1118	0.1177	0.4199
gpt2_prefix4	0.3101	0.0638	0.1307	0.2327	0.1020	0.1004	0.1012	0.4221

```
T5 -- Architecture Ablations
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tag	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	SPICE	SPIDEr	FENSE
t5	**0.3003 ±0.0122**	0.0574 ±0.0048	**0.1162 ±0.0036**	0.2275 ±0.0027	0.0847 ±0.0181	0.0866 ±0.0124	0.0857 ±0.0149	0.3954 ±0.0345
t5_depth1	0.2947	0.0546	0.1135	0.2267	0.0893	0.0953	0.0923	0.3755
t5_depth3	0.2400	0.0549	0.1044	0.2111	0.0968	0.0963	0.0965	0.4500
t5_drop0.1	0.2828	0.0570	0.1154	**0.2291**	**0.1308**	0.1044	0.1176	0.4389
t5_drop0.5	0.2877	0.0543	0.1143	0.2255	0.0971	0.0999	0.0985	0.3725
t5_prefix16	0.2508	0.0610	0.1095	0.2191	0.1263	0.1105	0.1184	**0.4947**
t5_prefix4	0.2683	**0.0655**	0.1157	0.2271	0.1240	**0.1139**	**0.1190**	0.4908

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LLAMA -- Architecture Ablations
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tag	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	SPICE	SPIDEr	FENSE
llama	**0.3104**	0.0669	**0.1310**	**0.2365**	0.1170	0.1156	0.1163	0.3626
llama_drop0.1	0.3046	0.0568	0.1282	0.2320	0.1019	0.1040	0.1030	**0.4065**
llama_drop0.5	0.2912	0.0602	0.1279	0.2227	0.1074	**0.1166**	0.1120	0.4060
llama_prefix16	0.2610	0.0517	0.1216	0.2105	0.0820	0.1138	0.0979	0.4045
llama_prefix4	0.2987	**0.0688**	0.1252	0.2286	**0.1333**	0.1159	**0.1246**	0.3838

4. Decoding Ablations (Test-Set Metrics)

Eval-only on baseline checkpoint -- no retraining. Top 15 configs per model by FENSE. Baseline row shows mean +/- std. **Bold** = best per column.

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GPT2 -- Decoding Ablations (top 15 by FENSE)
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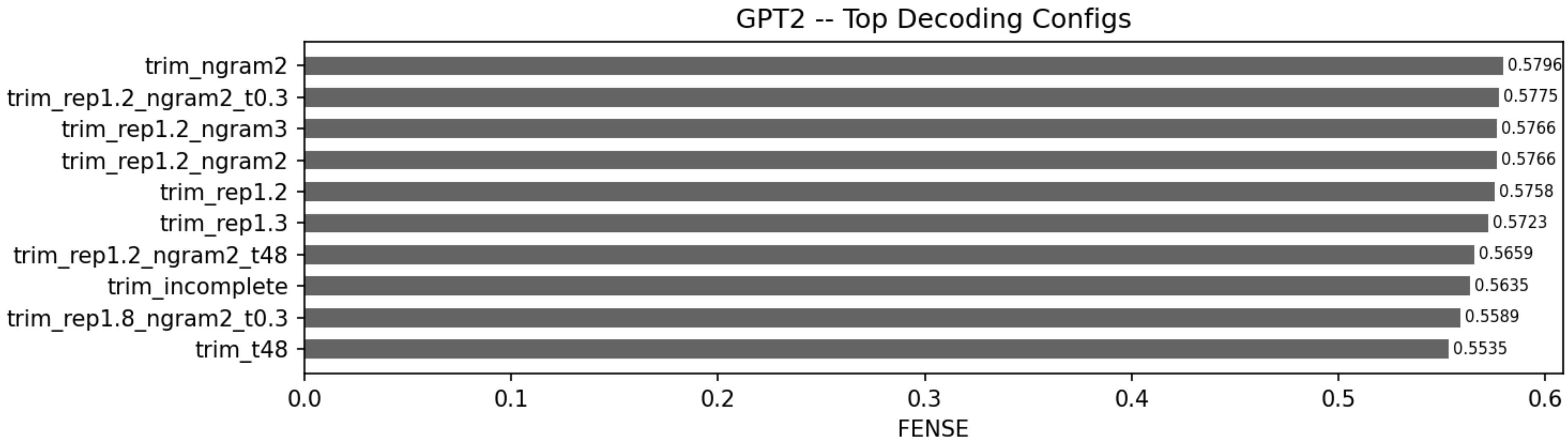
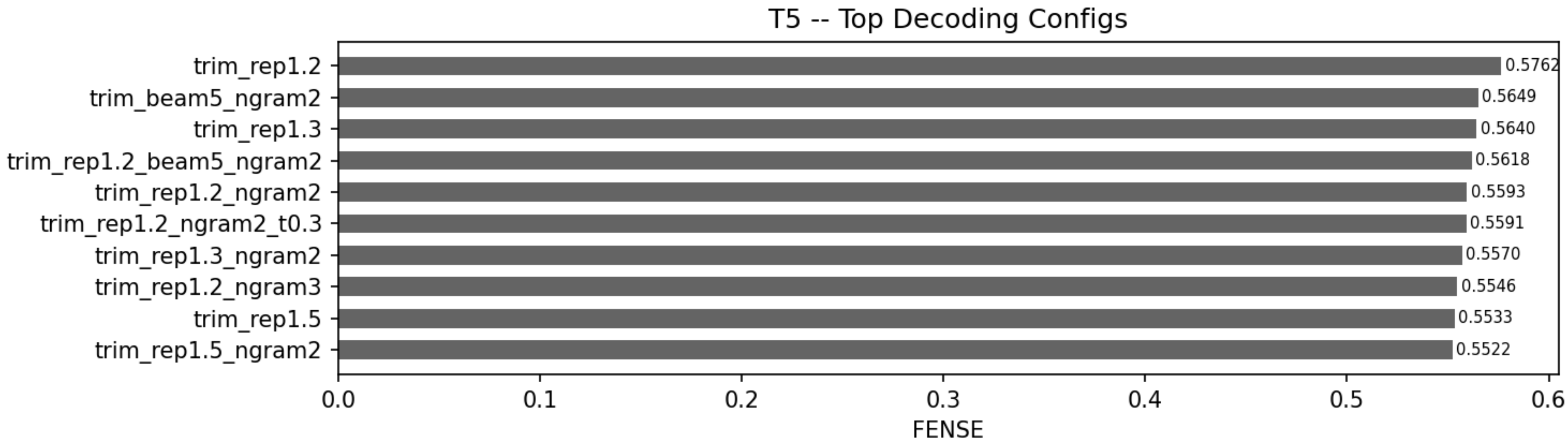
tag	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	SPICE	SPIDEr	FENSE
trim_ngram2	**0.3171**	0.0575	0.1210	0.2205	0.1092	**0.1081**	0.1087	**0.5796**
trim_rep1.2_ngram2_t0.3	0.2829	0.0454	0.1137	0.1970	0.1003	0.1062	0.1033	0.5775
trim_rep1.2_ngram3	0.2839	0.0407	0.1127	0.1944	0.0926	0.1045	0.0986	0.5766
trim_rep1.2_ngram2	0.2810	0.0409	0.1124	0.1931	0.0959	0.1042	0.1000	0.5766
trim_rep1.2	0.2834	0.0408	0.1124	0.1945	0.0924	0.1043	0.0983	0.5758
trim_rep1.3	0.2605	0.0374	0.1058	0.1843	0.0885	0.0990	0.0938	0.5723
trim_rep1.2_ngram2_t48	0.2011	0.0308	0.0938	0.1835	0.0692	0.1002	0.0847	0.5659
trim_incomplete	0.3091	**0.0605**	0.1202	**0.2303**	**0.1106**	0.1071	**0.1088**	0.5635
trim_rep1.8_ngram2_t0.3	0.2252	0.0348	0.0964	0.1708	0.0725	0.0940	0.0832	0.5589
trim_t48	0.2352	0.0433	0.1028	0.2140	0.0911	0.1010	0.0960	0.5535
rep1.2_ngram3	0.2869	0.0415	0.1212	0.1944	0.0900	0.1046	0.0973	0.4586
rep_1.2	0.2863	0.0417	**0.1214**	0.1952	0.0890	0.1056	0.0973	0.4552
rep_1.3	0.2794	0.0385	0.1188	0.1867	0.0842	0.1025	0.0934	0.4548
rep1.8_ngram2_t0.3	0.2723	0.0381	0.1185	0.1793	0.0920	0.1041	0.0980	0.4534
rep_1.8	0.2691	0.0322	0.1162	0.1763	0.0754	0.1001	0.0877	0.4488

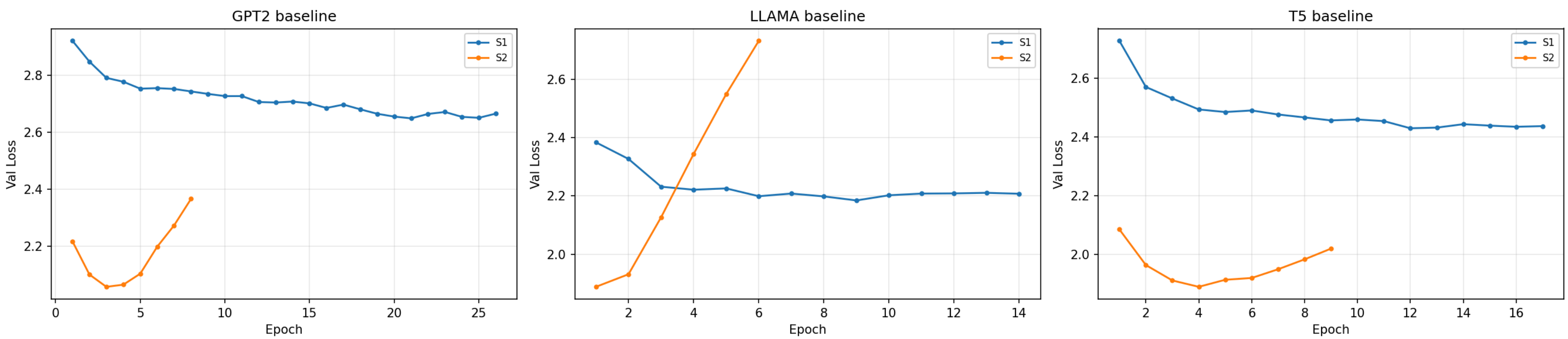


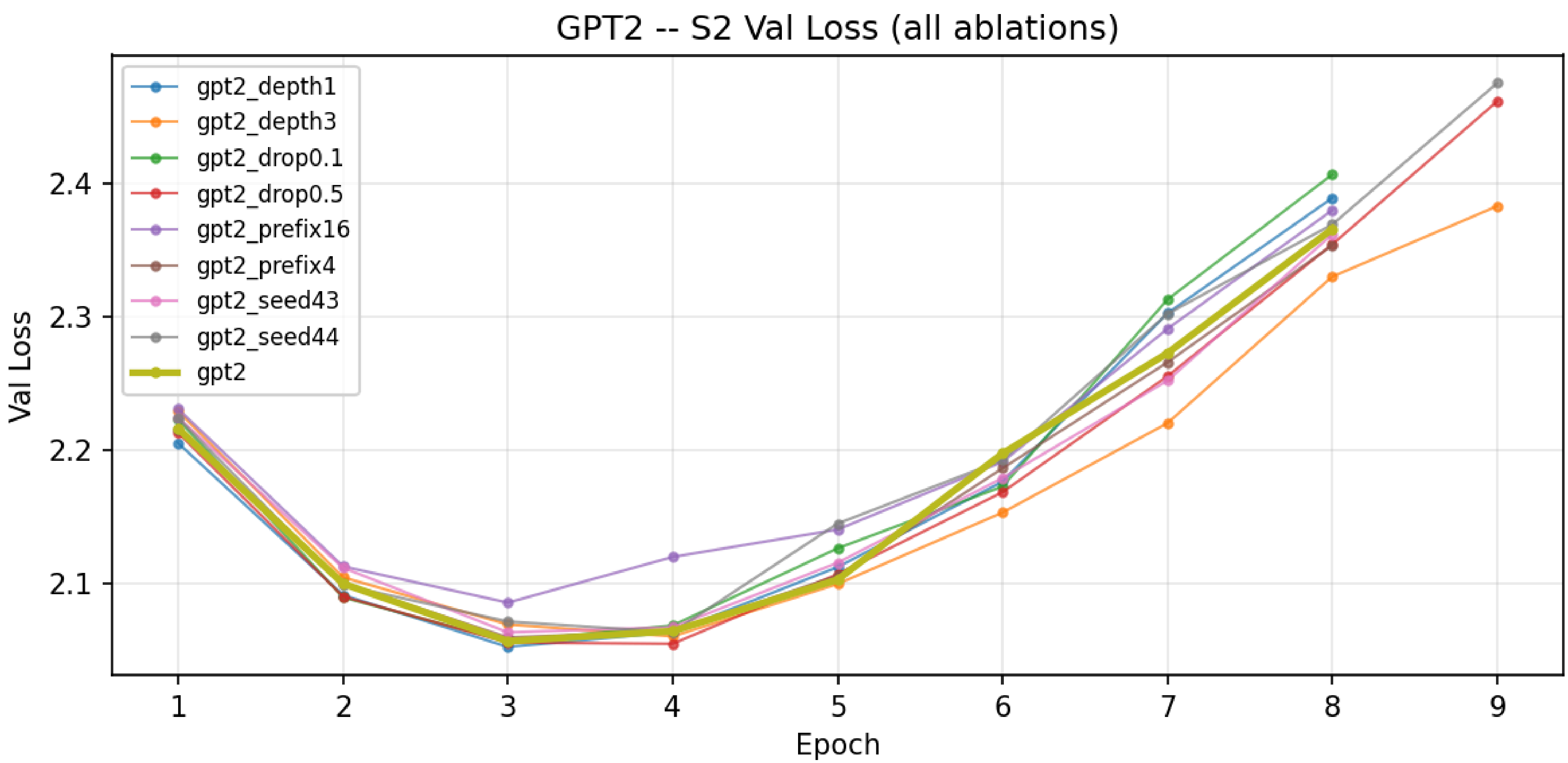
T5 -- Decoding Ablations (top 15 by FENSE)

tag	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	SPICE	SPIDEr	FENSE
trim_rep1.2	**0.3045**	0.0498	**0.1158**	**0.2203**	0.0863	0.0921	0.0892	**0.5762**
trim_beam5_ngram2	0.1923	0.0428	0.0918	0.1967	0.0878	0.1013	0.0945	0.5649
trim_rep1.3	0.2934	0.0424	0.1124	0.2065	0.0779	0.0987	0.0883	0.5640
trim_rep1.2_beam5_ngram2	0.1956	0.0442	0.0941	0.1949	0.0985	0.1023	0.1004	0.5618
trim_rep1.2_ngram2	0.2527	0.0296	0.0982	0.1904	0.0662	0.1000	0.0831	0.5593
trim_rep1.2_ngram2_t0.3	0.2500	0.0331	0.0978	0.1897	0.0611	0.0973	0.0792	0.5591
trim_rep1.3_ngram2	0.2629	0.0331	0.1018	0.1926	0.0711	0.1019	0.0865	0.5570
trim_rep1.2_ngram3	0.2725	0.0371	0.1017	0.2063	0.0738	0.0903	0.0821	0.5546
trim_rep1.5	0.2579	0.0264	0.1010	0.1817	0.0653	0.0916	0.0785	0.5533
trim_rep1.5_ngram2	0.2560	0.0261	0.1008	0.1803	0.0647	0.0918	0.0782	0.5522
trim_rep1.3_beam5	0.2262	**0.0563**	0.1041	0.2116	**0.1512**	**0.1099**	**0.1306**	0.5500
trim_ngram2	0.2406	0.0357	0.0969	0.1966	0.0569	0.0902	0.0736	0.5470
trim_rep1.2_ngram2_t48	0.1981	0.0226	0.0867	0.1831	0.0474	0.0942	0.0708	0.5462
trim_rep1.2_beam5_lp0.6	0.2236	0.0553	0.1034	0.2122	0.1396	0.1087	0.1241	0.5443
trim_incomplete	0.2670	0.0476	0.1091	0.2180	0.0662	0.0854	0.0758	0.5417

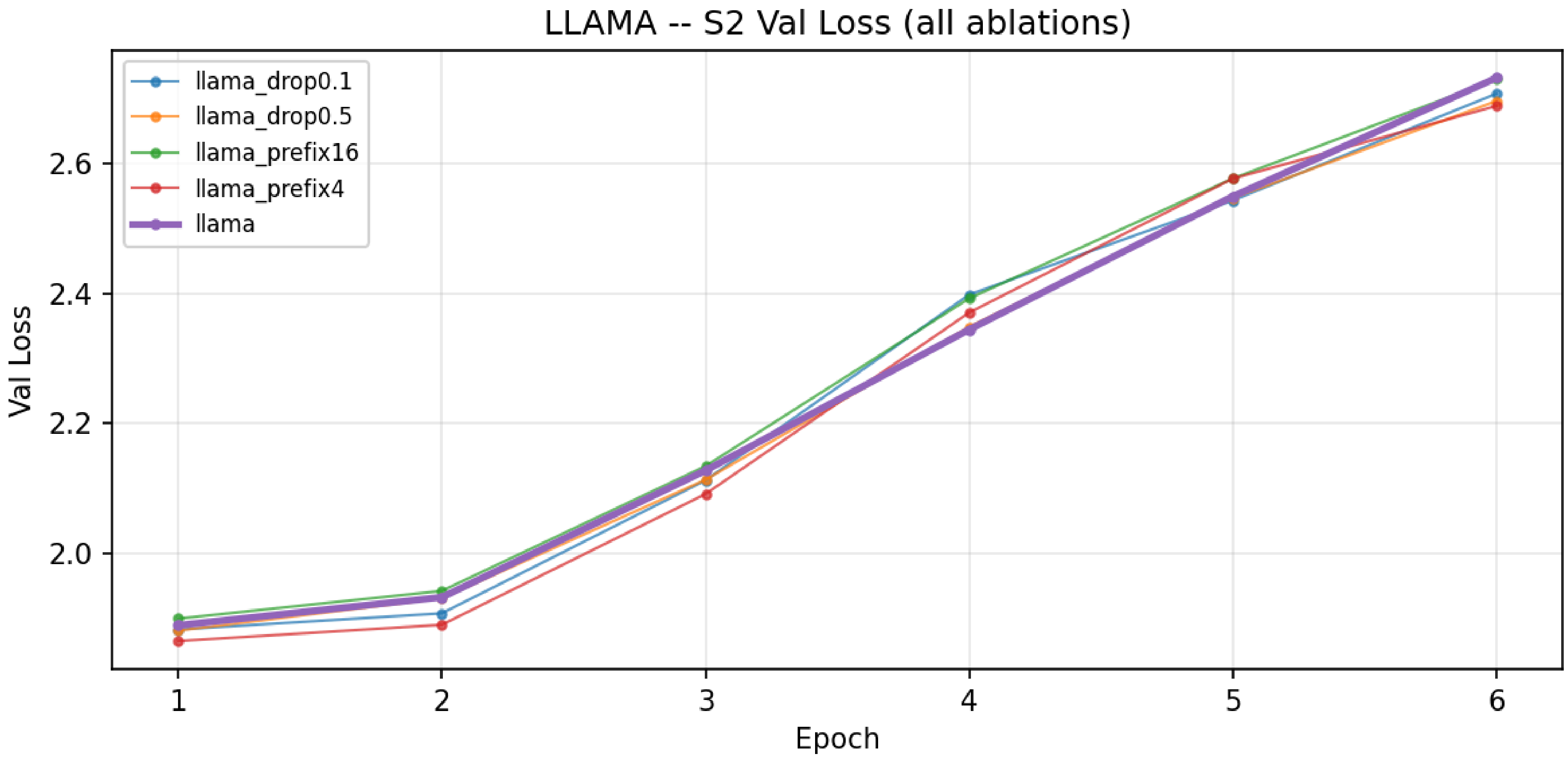


5. Training Curves

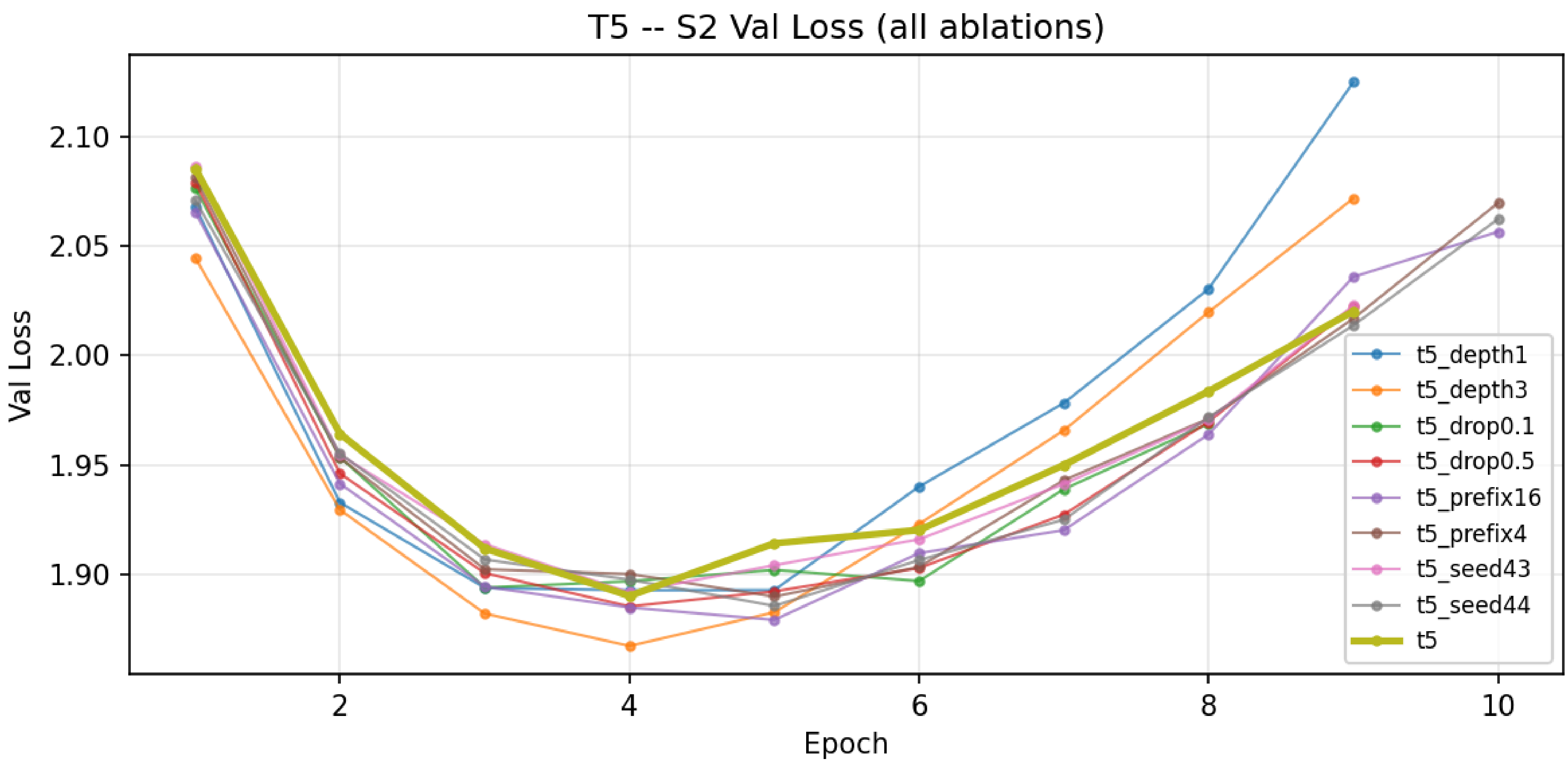




GPT2 -- S2 Val Loss (all ablations)
gpt2_depth1
gpt2_depth3
gpt2_drop0.1
gpt2_drop0.5
gpt2_prefix16
gpt2_prefix4
gpt2_seed43
gpt2_seed44
gpt2
Val Loss
2.1
2.2
2.3
2.4
Epoch
1
2
3
4
5
6
7
8
9



LLAMA -- S2 Val Loss (all ablations)
llama_drop0.1
llama_drop0.5
llama_prefix16
llama_prefix4
llama
Val Loss
2.0
2.2
2.4
2.6
1
2
3
4
5
6
Epoch



6. Sample Predictions (Best / Median / Worst by FENSE)

Loading weights: 100% 103/103 [00:00<00:00, 16131.94it/s]

Loading weights: 100% 199/199 [00:00<00:00, 19371.65it/s]

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============================================================
  GPT2 -- Sample Predictions (529 total)
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--- BEST (top 3) ---
  FENSE: 0.9183
  REF: A male vocalist sings this spirited song. The tempo is fast with intense drumming, loud harmonic electric guita
r feedback, bass guitar , emphatic vocals and keyboard accompaniment. The song is a loud, progressive, powerful youthf
ul and emphatic Hard Rock song.
  HYP: A male vocalist sings this spirited Rock song. The tempo is medium fast with keyboard harmony, steady drumming,
groovy bass lines, keyboard accompaniment and vocal backup. The song is youthful, energetic, enthusiastic, enthusiasti
c, enthusiastic and electrifying. This song is a Hard Rock/Heavy Metal/Rock. This

  FENSE: 0.8787
  REF: The low quality recording features a rock song that consists of a melancholic male vocal singing over punchy ki
ck and snare hits, shimmering ride cymbals, groovy bass guitar and acoustic rhythm guitar. It sounds emotional and hea
rtfelt.
  HYP: The low quality recording features a rock song that consists of passionate male vocal singing over groovy bass 
guitar, shimmering hi hats, punchy snare, punchy kick and snare hits, shimmering open hats, groovy bass guitar and wid
e electric guitar melody. It sounds passionate, emotional and the recording is noisy

  FENSE: 0.8653
  REF: The low quality recording features a live performance of a rock song and it features a passionate male vocal si
nging over groovy double bass, acoustic rhythm guitar, plucked violin melody and simple piano chords. It sounds reverb
erant, emotional, groovy and smooth - like something you would sing along to at the concert.
  HYP: The low quality recording features a live performance of a rock song that consists of passionate male vocal sin
ging over acoustic rhythm guitar, groovy bass guitar, shimmering shakers, punchy kick and snare hits, shimmering hi ha
ts and groovy bass guitar. It sounds passionate, emotional and the recording is noisy and in


--- MEDIAN ---
  FENSE: 0.4987
  REF: The song is an instrumental. The tempo is medium with an accordion player, guitar rhythm, flute player and othe
r percussion playing in harmony. The song is groovy and has gypsy flavour to it. The song audio quality is poor with a
mbient crowd noises.
  HYP: This is a live performance of a folk music piece. There is a clarinet playing the main melody while the upright
bass is playing in the background. The rhythmic background consists of a fast tempo acoustic drum beat. The atmosphere
is lively. This piece could be used in the background of a folk dance course. It could

  FENSE: 0.4976
  REF: The low quality recording features a shofar repetitive melody. The recording is in mono and so noisy that it sl
ightly distorts the audio. It sounds soulful and reverberant.
  HYP: This is a live recording of a shofar being played. The shofar is being played in a high-pitched manner. There i
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s a high-pitched sound in the recording. This piece could be used in the soundtracks of a historical drama during the
scenes of a victory parade. It could also

  FENSE: 0.4975
  REF: The low quality recording features a tutorial that consists of a french horn melody played in-between flat male
vocal talking. The recording is noisy, mono and quiet.
  HYP: This is a live recording of a trumpet solo. The trumpet is playing a simple melody in a high key. The brass sec
tion is playing a simple tune. The atmosphere is epic. This piece could be used in the soundtrack of a historical dram
a during the scenes of a victory parade. It could also be used in the soundtrack


--- WORST (bottom 3) ---
  FENSE: 0.0348
  REF: This is the recording of a tuning video in the key of D sharp. There is an electric guitar playing the note on
each string repeatedly. This piece could be used to lift samples to be used in a beat or in the making of a virtual gu
itar plug-in.
  HYP: This music is instrumental. The tempo is medium with a stringed instrument playing a simple melody with a lot o
f vibrato. The music is simple, simple, mellow, simple, simple, simple and simple. This music is a stringed instrument
instrumental. The music is simple, simple, simple, simple and simple

  FENSE: 0.0336
  REF: The low quality recording features a compilation of a few different recordings that mostly consist of female vo
cals talking with each other and a loud crowd talking in the background. However, in the first section, there are also
some claps and female vocals mumbling a song. The arrangement of the recordings is weird, but in context it probably m
akes sense.
  HYP: A female vocalist sings this spirited song. The tempo is medium fast with keyboard harmony, steady drumming, st
eady bass line and vocal backup. The song is a spirited, enthusiastic, enthusiastic, enthusiastic, enthusiastic, enthu
siastic, enthusiastic, enthusiastic and youthful Pop. This song is a Pop/R&B. This song

  FENSE: 0.0233
  REF: This clip features noises created by mixing output from different effect pedals. This is a synth noise. There a
re no instruments or voices in this song.
  HYP: This is a live recording of a didgeridoo being played. The didgeridoo is being played at a high pitch. There is
a high-pitched high-pitched sound that can be heard. The didgeridoo is being played at a medium-to-high pitch. There i
s a
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Loading weights: 100% 103/103 [00:00<00:00, 13666.12it/s]

Loading weights: 100% 199/199 [00:00<00:00, 15512.52it/s]

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============================================================
  T5 -- Sample Predictions (529 total)
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--- BEST (top 3) ---
  FENSE: 0.8309
  REF: This is a meditation music piece. There is an ambient synth sound playing very faint notes. There are natural s
ounds that can be heard in the piece such as the water flowing and the crickets chirping. The atmosphere of the piece
is relaxing and soothing. The piece would suit perfectly as background music for a meditation video. It could also be
playing in the background at spas and wellness centers.
  HYP: This is a meditation music piece. There is a mellow harp playing the main tune. The atmosphere is relaxing. Thi
s piece could be used in meditation videos. It could also be used in meditation videos.

  FENSE: 0.7984
  REF: This is an experimental rock/indie rock music piece. The rhythmic background consists of a peculiar 6/8 acousti
c drum beat that has shifted kick drum hits. There are sleigh bells accompanying the drums. The synth bass is playing
a simple and repeated theme. The atmosphere is eerie. This piece could be used in the soundtrack of a noir movie where
a character is devising a plan to deal with conflict.
  HYP: This is a groovy electronic music piece. There is a groovy bassline being played. The rhythmic background consi
sts of a fast-paced electronic drum beat. The atmosphere is energetic. This piece could be used in the soundtrack of a
video game.

  FENSE: 0.7950
  REF: This is a dancehall music piece. There is a male vocalist singing with a Jamaican accent. An electric guitar an
d a bass guitar are playing the melody in the background while occasional keyboard notes can be heard. The rhythmic ba
ckground is provided by the mild tempo basic 4/4 played on the acoustic drums. There is an echo delay effect on the sn
are drum every now and then. The atmosphere of the piece is chill and groovy. This piece could be playing through loud
speakers at a beach. It could also suit well at a summer barbecue party.
  HYP: This is a funk/funk music piece. There is a male vocalist singing melodically. The melody is being played by th
e keyboard while the rhythmic background consists of a simple electronic drum beat. The atmosphere is lively. This pie
ce could be used in the background of electronic dance music


--- MEDIAN ---
  FENSE: 0.4878
  REF: A male ukulele player plays this emotional solo. The song is medium tempo with no other instrumentation than an
ukulele playing. The song is emotional and poignant. The song is an instrumental tune.
  HYP: This is a live recording of a saxophone solo being played. There is a saxophone playing a solo. The tempo is sl
ow with the recording is of poor quality. The music is a live recording and the audio quality is poor.

  FENSE: 0.4875
  REF: instrumental rock music with a drum kit, lots of cymbal crashes, acoustic guitar strumming, harmonized lead gui
tars and an active bass guitar.
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  HYP: This is a rock music piece. There is a male vocalist singing melodically. The electric guitar is playing a simp
le tune with a bass guitar playing in the background. The rhythmic background consists of a fast-paced acoustic drum b
eat. The atmosphere is energetic

  FENSE: 0.4867
  REF: The low quality recording features a children's song played on TV that consists of funny male and female vocal
singing over mellow piano chord progression and happy sounding bells melody. It is really noisy and in mono, but it so
unds happy and fun - like something kids would listen to.
  HYP: This is a live recording of a folk music piece. There is a male vocalist singing melodically. The melody is bei
ng played by the acoustic guitar while the rhythm is being played by the acoustic guitar. The atmosphere is joyful. Th
is piece could be used


--- WORST (bottom 3) ---
  FENSE: 0.0281
  REF: This music is an enthralling Sitar instrumental. The tempo is fast with an animated and melodic Sitar lead Harm
ony and tanpura accompaniment. The music is based on Hindustani Classical notes. It is captivating, intense, mellifluo
us, engaging,and fervent.
  HYP: This is a live recording of a shofar being played. There is a shofar being played. There is a shofar being play
ed. There is a shofar being played. There is a shofar being played. There is a shofar being played

  FENSE: 0.0279
  REF: Here we have an energetic, up-tempo japanese rock song. The drumming is potent, punchy and predominant. The dru
mmer plays the ride cymbal frequently, creating a shimmery effect in the song. The female singer has a calming voice.
  HYP: This is a live recording of a jazz music piece. There is a male vocalist singing melodically in the lead. The m
elody is being played by the acoustic guitar while the bass guitar is playing in the background. The rhythm is provide
d by a simple acous

  FENSE: 0.0273
  REF: There is a high-pitched computer-like sound effect in the piece. It gives this cold, uneasy feeling. It could b
e used in a cyberpunk video game or a dystopian movie.
  HYP: This song contains digital drums playing a simple groove with a lot of reverb on it. A synth bass is playing a
melody along with a synth bass playing a simple bassline. A synth sound is playing a melody in the mid range. This son
g may be playing in a
```

Loading weights: 100% 103/103 [00:00<00:00, 13760.14it/s]

Loading weights: 100% 199/199 [00:00<00:00, 16133.81it/s]

```
================================================================
  LLAMA -- Sample Predictions (529 total)
================================================================

--- BEST (top 3) ---
  FENSE: 0.8896
  REF: This is an instrumental heavy metal piece. The distorted electric guitar is playing an aggressive riff while ac
companied by a bass guitar. In the rhythmic background, there is a slow but hard-hitting heavy metal beat played on th
e acoustic drums. The atmosphere is loud and violent. This piece can be used in the soundtrack of a movie/TV show invo
lving crime. It could also be played at a rock bar or a sports venue.
  HYP: This is a hardcore metal music piece. There is a distorted electric guitar playing the main tune while a bass g
uitar is playing in the background. The rhythm is provided by a fast-paced metal drum beat. The atmosphere is aggressi
ve and energetic. This piece could be used in the soundtrack of a hardcore metal movie. It could

  FENSE: 0.8841
  REF: The low quality recording features a live performance of a jazz song that consists of a tenor sax solo melody p
laying over mellow piano chords, sustained strings, groovy bass, snappy rimshots, shimmering hi hats and punchy kicks.
It sounds emotional, passionate and soulful.
  HYP: The low quality recording features a live performance of a jazz song that consists of a saxophone solo melody o
ver groovy double bass, groovy piano chords and shimmering shakers. It sounds passionate, emotional and the recording
is noisy and in mono. The recording is a bit noisy and in mono. It sounds like

  FENSE: 0.8772
  REF: This is an instrumental punk rock/dub music piece. There is a distorted electric playing the main tune with a b
ass guitar in the background. The rhythmic background consists of a loud and rapid acoustic drum beat. The atmosphere
is aggressive. This piece could be used in the soundtrack of a violent video game.
  HYP: This is a hardcore rock music piece. There is a loud electric guitar playing the main tune while a bass guitar
is playing in the background. The rhythm is provided by a loud rock drum beat. The atmosphere is energetic and rebelli
ous. This piece could be used in the soundtrack of a hardcore rock movie. It could


--- MEDIAN ---
  FENSE: 0.4671
  REF: The first sound effect in this clip is the sound of old TV static. Then we hear a retro video game sound used t
o illustrate when the character jumps up. Then we have a sustained calming sound which is like the  sound after a larg
e bell has been rung.
  HYP: This is a compilation of various sound effects. There is a sound effect that sounds like a camera shuttering. T
here is a sound effect that sounds like a camera being turned on. There is a sound effect that sounds like a camera is
being turned off. There is a sound effect that sounds like a camera is being

  FENSE: 0.4664
  REF: A hard digital kick is playing along with a monotone bass sound. A malled instrument is playing a repetitive me
lody in the mid to high register. Evolving liquid synth sounds are rising into focus and up in pitch then a male voice
```

```
sample comes in before the drum breaks into a new part with a snare sound. This may be playing at a rave in the woods.
  HYP: This is a techno remix of a pop music piece. There is a female vocal sample in the beginning of the piece. The
rhythmic background consists of a techno drum beat. The atmosphere is energetic and the piece has a strong dance groov
e. This piece could be used in the soundtrack of a dance movie or a dance

  FENSE: 0.4629
  REF: An accordion is playing short notes on the offbeat along with soft snare hits being played with brushes. A bass
is playing straight, changing the note on every beat. Another accordion is answering the yodeling male voice with a sh
ort melody. The male voice is loud, clear and sings a melody in the higher register. A guitar-like instrument is strum
ming chords in the higher register. The instruments are panned to the left and right side of the speakers. This song m
ay be playing at a concert in the countryside.
  HYP: This is a traditional middle eastern song. There is a male vocalist singing melodically in the lead. The melody
is being played by the string instruments and the flute. The rhythmic background consists of a simple percussion beat.
The atmosphere is vibrant and lively. This song would fit perfectly in the background at a middle eastern


--- WORST (bottom 3) ---
  FENSE: 0.0307
  REF: A male vocalist sings this sad melody. The tempo is  slow with an acoustic guitar accompaniment and a droning k
eyboard harmony. The song is soft ,mellow, sad, emotional, sentimental, regretful , lonely and poignant. This song is
an Alternative Rock/Indie.
  HYP: This is a live performance of a traditional chinese opera. There is a female vocalist singing melodically in th
e lead. The melody is being played by the zheng, the pipa and the guzheng. The atmosphere is calming and relaxing. Thi
s piece could be used in the soundtrack of a drama movie or a

  FENSE: 0.0288
  REF: A digital drum is playing a simple rhythm along with a synth bassline. A very pregnant synth lead i s playing a
catchy and repeating melody in the higher register. In the background you can hear feet stumping noises. This song may
be playing in a club.
  HYP: This is the live recording of a traditional south asian folk music piece. There is a male vocalist singing melo
dically in the lead. The rhythmic background consists of a tabla playing the percussion. The atmosphere is lively. Thi
s piece could be used in the soundtrack of a Bollywood movie. It could also be used in the

  FENSE: 0.0287
  REF: Tablas and accordion lead a coed group chant with a male lead singer. Sounds spiritual.
  HYP: A female vocalist sings this soprano melody. The tempo is fast with a lively orchestra accompaniment. The song
is a lively, vibrant, vibrant, vibrant, vibrant, and engaging. This song is a Classical Opera. This song is a Classica
l Opera. This song is a Classical Opera. This song is a
```

7. Failure Cases

```
t5/t5_depth3: 4 failures out of 529
  [word repetition]
  REF: The excerpt starts off with 3 bass drum beats after which an explosion and a crumbling sound can be
  HYP: The low quality recording features a wide wide wide wide wide wide reverberant snare, wide wide tinn

  [word repetition]
  REF: This clip features noises created by mixing output from different effect pedals. This is a synth noi
  HYP: This music is instrumental. The tempo is fast with a droning droning harmony of the droning droning

  [word repetition]
  REF: A male vocalist sings this vigorous heavy metal song. The tempo is fast with intense  electric guita
  HYP: The low quality recording features a metal song that consists of a passionate male vocal singing ove

  ... and 1 more

llama/llama_drop0.5: 1 failures out of 529
  [word repetition]
  REF: The music is an instrumental gameplay track. The track has sounds of growling monsters, church bells
  HYP: This is a video game theme. There is a male voice narrating the video game. The sound quality is poo

llama/llama_prefix4: 1 failures out of 529
  [repetitive]
  REF: The low quality recording features a steel drums melody played over punchy kick and snare hits, shim
  HYP: The low quality recording features a steel steel steel steel steel steel steel steel steel steel ste
```

8. Full Summary (All Ablations, Sorted by FENSE)

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GPT2 -- All Ablations (41 configs)
```

tag	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	SPICE	SPIDEr	FI
trim_ngram2	**0.3171**	0.0575	0.1210	0.2205	0.1092	0.1081	0.1087	**0.**
trim_rep1.2_ngram2_t0.3	0.2829	0.0454	0.1137	0.1970	0.1003	0.1062	0.1033	0.
trim_rep1.2_ngram3	0.2839	0.0407	0.1127	0.1944	0.0926	0.1045	0.0986	0.
trim_rep1.2_ngram2	0.2810	0.0409	0.1124	0.1931	0.0959	0.1042	0.1000	0.
trim_rep1.2	0.2834	0.0408	0.1124	0.1945	0.0924	0.1043	0.0983	0.
trim_rep1.3	0.2605	0.0374	0.1058	0.1843	0.0885	0.0990	0.0938	0.
trim_rep1.2_ngram2_t48	0.2011	0.0308	0.0938	0.1835	0.0692	0.1002	0.0847	0.
trim_incomplete	0.3091	0.0605	0.1202	0.2303	0.1106	0.1071	0.1088	0.
trim_rep1.8_ngram2_t0.3	0.2252	0.0348	0.0964	0.1708	0.0725	0.0940	0.0832	0.
trim_t48	0.2352	0.0433	0.1028	0.2140	0.0911	0.1010	0.0960	0.
rep1.2_ngram3	0.2869	0.0415	0.1212	0.1944	0.0900	0.1046	0.0973	0.
rep_1.2	0.2863	0.0417	0.1214	0.1952	0.0890	0.1056	0.0973	0.
rep_1.3	0.2794	0.0385	0.1188	0.1867	0.0842	0.1025	0.0934	0.
rep1.8_ngram2_t0.3	0.2723	0.0381	0.1185	0.1793	0.0920	0.1041	0.0980	0.
rep_1.8	0.2691	0.0322	0.1162	0.1763	0.0754	0.1001	0.0877	0.
rep_1.5	0.2744	0.0356	0.1178	0.1816	0.0801	0.1012	0.0906	0.
tokens_48	0.2775	0.0529	0.1120	0.2253	0.1039	0.1053	0.1046	0.
ngram_2	0.3123	0.0568	0.1277	0.2200	0.1122	0.1066	0.1094	0.
gpt2_drop0.5	0.3052	0.0653	0.1297	0.2317	0.1080	0.1050	0.1065	0.
ngram_3	0.3131	0.0624	0.1277	0.2315	0.1116	0.1064	0.1090	0.
gpt2_prefix4	0.3101	0.0638	0.1307	0.2327	0.1020	0.1004	0.1012	0.
gpt2_prefix16	0.3093	0.0698	0.1324	0.2345	0.1235	0.1118	0.1177	0.
sample_t0.9_p0.95	0.2997	0.0571	0.1249	0.2202	0.0898	0.1067	0.0982	0.
ngram_4	0.3114	0.0631	0.1271	0.2334	0.1114	0.1067	0.1091	0.
gpt2_seed43	0.3126	**0.0702**	**0.1338**	0.2361	0.1254	0.1027	0.1141	0.
gpt2_drop0.1	0.3106	0.0696	0.1337	**0.2371**	**0.1268**	0.1152	**0.1210**	0.
sample_t1.0_p0.95	0.2943	0.0536	0.1223	0.2145	0.0817	0.1025	0.0921	0.
sample_t0.3_p0.8	0.3134	0.0690	0.1299	0.2345	0.1202	0.1105	0.1153	0.
rep_1.1	0.3029	0.0494	0.1261	0.2138	0.0978	0.1014	0.0996	0.
gpt2_depth3	0.2951	0.0600	0.1255	0.2201	0.0952	0.1000	0.0976	0.
gpt2	0.3091 ±0.0019	0.0631 ±0.0039	0.1265 ±0.0041	0.2329 ±0.0017	0.1109 ±0.0076	0.1091 ±0.0032	0.1100 ±0.0024	0.3998 ±0.

tag	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	SPICE	SPIDEr	FI
gpt2_depth1	0.3103	0.0631	0.1288	0.2339	0.1078	0.1164	0.1121	0.
tokens_96	0.2205	0.0442	0.1300	0.2076	0.0254	0.1050	0.0652	0.
tokens_32	0.1754	0.0331	0.0909	0.2054	0.0589	0.0942	0.0765	0.
sample_t0.7_p0.9	0.3026	0.0629	0.1267	0.2253	0.1005	0.1079	0.1042	0.
gpt2_seed44	0.3097	0.0692	0.1332	0.2356	0.1220	0.1064	0.1142	0.
beam_2	0.2990	0.0699	0.1263	0.2323	0.1217	**0.1182**	0.1200	0.
beam_5	0.2953	0.0692	0.1252	0.2295	0.1121	0.1114	0.1118	0.
beam4_lp0.6	0.2976	0.0702	0.1252	0.2323	0.1155	0.1123	0.1139	0.
beam4_lp1.5	0.2976	0.0702	0.1252	0.2323	0.1155	0.1123	0.1139	0.
beam_4	0.2976	0.0702	0.1252	0.2323	0.1155	0.1123	0.1139	0.

T5 -- All Ablations (52 configs)

tag	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	SPICE	SPIDEr
trim_rep1.2	0.3045	0.0498	0.1158	0.2203	0.0863	0.0921	0.0892
trim_beam5_ngram2	0.1923	0.0428	0.0918	0.1967	0.0878	0.1013	0.0945
trim_rep1.3	0.2934	0.0424	0.1124	0.2065	0.0779	0.0987	0.0883
trim_rep1.2_beam5_ngram2	0.1956	0.0442	0.0941	0.1949	0.0985	0.1023	0.1004
trim_rep1.2_ngram2	0.2527	0.0296	0.0982	0.1904	0.0662	0.1000	0.0831
trim_rep1.2_ngram2_t0.3	0.2500	0.0331	0.0978	0.1897	0.0611	0.0973	0.0792
trim_rep1.3_ngram2	0.2629	0.0331	0.1018	0.1926	0.0711	0.1019	0.0865
trim_rep1.2_ngram3	0.2725	0.0371	0.1017	0.2063	0.0738	0.0903	0.0821
trim_rep1.5	0.2579	0.0264	0.1010	0.1817	0.0653	0.0916	0.0785
trim_rep1.5_ngram2	0.2560	0.0261	0.1008	0.1803	0.0647	0.0918	0.0782
trim_rep1.3_beam5	0.2262	0.0563	0.1041	0.2116	0.1512	0.1099	0.1306
trim_ngram2	0.2406	0.0357	0.0969	0.1966	0.0569	0.0902	0.0736
trim_rep1.2_ngram2_t48	0.1981	0.0226	0.0867	0.1831	0.0474	0.0942	0.0708
trim_rep1.2_beam5_lp0.6	0.2236	0.0553	0.1034	0.2122	0.1396	0.1087	0.1241
trim_incomplete	0.2670	0.0476	0.1091	0.2180	0.0662	0.0854	0.0758
trim_beam5_lp0.6	0.2197	0.0549	0.1024	0.2119	0.1412	0.1086	0.1249
trim_rep1.2_beam5	0.2271	0.0567	0.1042	0.2131	0.1519	0.1112	0.1316
trim_beam5	0.2257	0.0567	0.1038	0.2137	0.1535	0.1105	0.1320
trim_beam5_lp1.5	0.2316	0.0582	0.1048	0.2125	0.1516	0.1103	0.1310
trim_rep1.8_ngram2_t0.3	0.1682	0.0158	0.0766	0.1419	0.0352	0.0747	0.0550
trim_t48	0.1647	0.0292	0.0858	0.1969	0.0401	0.0797	0.0599
rep_1.5	0.2793	0.0283	0.1062	0.1834	0.0750	0.0926	0.0838
ngram_2	0.2510	0.0373	0.0992	0.1986	0.0614	0.0914	0.0764
t5_prefix16	0.2508	0.0610	0.1095	0.2191	0.1263	0.1105	0.1184
rep1.8_ngram2_t0.3	0.2566	0.0225	0.1025	0.1583	0.0650	0.0917	0.0783
rep1.2_ngram3	0.2911	0.0406	0.1056	0.2094	0.0768	0.0902	0.0835
t5_prefix4	0.2683	0.0655	0.1157	0.2271	0.1240	0.1139	0.1190
beam_2	0.2395	0.0591	0.1073	0.2156	0.1226	**0.1159**	0.1193
rep_1.8	0.2620	0.0222	0.1042	0.1611	0.0672	0.0924	0.0798
beam_5	0.2361	0.0587	0.1061	0.2177	**0.1609**	0.1102	**0.1356**
beam4_lp0.6	0.2242	0.0564	0.1032	0.2129	0.1411	0.1077	0.1244

tag	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	SPICE	SPIDEr	
beam4_lp1.5	0.2342	0.0590	0.1053	0.2137	0.1519	0.1088	0.1303	
beam_4	0.2299	0.0579	0.1044	0.2149	0.1541	0.1085	0.1313	
rep_1.3	0.3193	0.0481	0.1187	0.2102	0.0871	0.0998	0.0935	
rep_1.1	**0.3251**	0.0586	0.1224	0.2291	0.0997	0.0915	0.0956	
sample_t1.0_p0.95	0.2573	0.0418	0.1042	0.2068	0.0790	0.0899	0.0844	
sample_t0.9_p0.95	0.2600	0.0452	0.1046	0.2071	0.0878	0.0994	0.0936	
t5_depth3	0.2400	0.0549	0.1044	0.2111	0.0968	0.0963	0.0965	
t5_seed44	0.2926	0.0651	0.1201	**0.2328**	0.1201	0.1108	0.1155	
sample_t0.7_p0.9	0.2670	0.0517	0.1074	0.2139	0.0985	0.0955	0.0970	
t5_drop0.1	0.2828	0.0570	0.1154	0.2291	0.1308	0.1044	0.1176	
rep_1.2	0.3222	0.0551	0.1206	0.2237	0.0869	0.0920	0.0894	
ngram_3	0.2808	0.0456	0.1073	0.2149	0.0816	0.0921	0.0869	
ngram_4	0.3021	0.0577	0.1171	0.2268	0.0844	0.0874	0.0859	
sample_t0.3_p0.8	0.2960	0.0572	0.1148	0.2259	0.0784	0.0890	0.0837	
t5	0.3003 ±0.0122	0.0574 ±0.0048	0.1162 ±0.0036	0.2275 ±0.0027	0.0847 ±0.0181	0.0866 ±0.0124	0.0857 ±0.0149	0.3954
tokens_96	0.3005	0.0575	0.1163	0.2277	0.0849	0.0869	0.0859	
tokens_48	0.2404	0.0425	0.1021	0.2152	0.0585	0.0872	0.0729	
t5_seed43	0.3164	**0.0663**	**0.1234**	0.2314	0.0961	0.1032	0.0996	
t5_depth1	0.2947	0.0546	0.1135	0.2267	0.0893	0.0953	0.0923	
t5_drop0.5	0.2877	0.0543	0.1143	0.2255	0.0971	0.0999	0.0985	
tokens_32	0.1306	0.0227	0.0792	0.1924	0.0275	0.0705	0.0490	

LLAMA -- All Ablations (5 configs)

tag	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr-D	SPICE	SPIDEr	FENSE
llama_drop0.1	0.3046	0.0568	0.1282	0.2320	0.1019	0.1040	0.1030	**0.4065**
llama_drop0.5	0.2912	0.0602	0.1279	0.2227	0.1074	**0.1166**	0.1120	0.4060
llama_prefix16	0.2610	0.0517	0.1216	0.2105	0.0820	0.1138	0.0979	0.4045
llama_prefix4	0.2987	**0.0688**	0.1252	0.2286	**0.1333**	0.1159	**0.1246**	0.3838
llama	**0.3104**	0.0669	**0.1310**	**0.2365**	0.1170	0.1156	0.1163	0.3626